

- 69** The table shows information about a human genetic condition called sickle cell anaemia and an infection called malaria. Both sickle cell anaemia and malaria can be fatal.

<i>genotype</i>	<i>phenotype</i>	<i>comment</i>
MM	does not show sickle cell anaemia	can be infected with malaria
Mm	does not show sickle cell anaemia	shows resistance to malaria
mm	shows sickle cell anaemia	shows more resistance to malaria than Mm

Which of the following statements is/are correct?

- 1** In areas without malaria, human populations are likely to have a low number of people with the m allele.
 - 2** In areas with malaria, only those individuals that are heterozygous will be able to pass on their alleles to the next generation.
 - 3** Presence of malaria has caused a mutation of the M allele to the m allele leading to an increased chance of survival in the heterozygous state.
- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3